Spain V20 | Complete archive restored

Additive restoration and corrected marine-fleet cost audit

What is preserved

- Original Spain V13 integrated archive — 201.38 MB, 428 entries, all art, code, figures and complete documents preserved in its unmodified ZIP.
- Original V17 Nature Sustainability archive — 9.56 MB, 46 entries, unmodified.
- Original V18 marine-hybrid and SMR archive — 12.63 MB, 68 entries, unmodified.
- All V19 documents, model, figures, baseline and SHA-256 manifests retained unchanged.

All 542 historic internal entries were verified against recorded SHA-256 hashes. ZIP CRC checks passed on the three originals and the outer package.

Correction: charge the solar-plus-kite fleet for its kite

The original V19 pv_kite option was credited with kite-induced fuel savings but omitted the cost of the kite module. V20 corrects that financial allocation without altering the older source archives.

Scenario	Fuel saved (TWh/yr)	Net annual saving (€M)
7-kn 100-y PV + kite, old V19	0.268	36.42 (superseded)
7-kn 100-y PV + kite, V20	0.268	29.86
6-kn 100-y existing kite, V20	0.336	36.31
4.5-kn 30-y full hybrid, V20	0.557	−21.72

A 100-year ship hull is an unvalidated sensitivity, not a demonstrated construction lifetime. The corrected model explores financing, replacement, maintenance and capital-premium ranges.

Harvesting, seeding and maintenance

The solar-charged utility drone performs initial seeding and condition-triggered spot replanting. The scenario does not require complete annual reseeding. Harvesters may operate in daylight or at night subject to wind, electricity storage and safe sea conditions.

Solar and wind on the 280 modeled harvesters deliver about 51.84 GWh/year versus 32.82 GWh/year for small cutting heads. This does not eliminate the separate 3.545 TWh/year mixed harvesting/support fleet energy allowance. Under an illustrative 15% work-avoidance case, renewable power and electrification reduce fleet-fuel demand by 0.570 TWh/year; 2.975 TWh/year remains.

Reproducibility

Inside V20_Corrected_Marine_Model, run python model_v20.py and python test_v20.py. The corrected outputs are the files prefixed V20_. All prior original ZIPs and the V19 historical analysis remain within this delivery.

This marine-fleet sensitivity does not constitute a complete national system rerun, field-tested 100-year vessel design, or confirmed round-the-clock renewable propulsion. Measurements and engineering validation remain necessary.